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Introduction to Predictive Analytics

UNIT - I

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Introduction to Analytics

Understanding the Art and Science of Data-Driven Decisions

Objective: To explore the foundational concepts of analytics, its real-world motivations, and how it differs from traditional business intelligence.

Engaging Motivation - A Marketing Dilemma

- **The Problem:** A direct response company wanted to sell books and DVDs via subscription but needed to know which customers were worth the significant marketing spend.
- **The Traditional Approach:** They relied on small test mailings to see if overall expected response rates justified a larger campaign.
- **The Analytical Solution:** An analyst realized they could identify key characteristics of responders and generate a mathematical score for each customer.

The Outcome: Using predictive analytics techniques, they determined the exact minimum score required to make the marketing campaign highly profitable.

Engaging Motivation - The Healthcare Challenge

- **The Problem:** A hospital wanted to improve early risk detection for patients using complex clinical decision models.
- **The Constraint:** The full statistical framework produced accurate results offline, but it was too slow and complicated for rapid bedside decision support.
- **The Analytical Solution:** Data scientists built predictive models that estimated patient risk using observable indicators such as age, vital signs, lab results, and medical history.

The Outcome: The risk scores were accurate enough to support timely intervention and better patient management without requiring excessive computation.

The Core Connection - Estimating the Unknown

- **A Shared Problem:** Predicting human purchasing behavior and identifying patient risk in healthcare seem entirely unrelated.
- **The Common Thread:** Both scenarios share a profound similarity: they need to estimate values that are unknown but tremendously useful.
- **The Analytics Function:** Whether predicting a human decision or a medical outcome, models predict values that are historically known but unknown at the exact moment a critical decision is needed.

Real-World Value: Countless businesses, hospitals, educational institutions, and government agencies use historic data every day to aid or fully automate these vital decisions.

Defining Analytics

- **Definition:** Analytics is the process of using computational methods to discover and report influential patterns in data.
- **The Primary Goal:** The ultimate goal of analytics is to gain insight and to actively affect decisions.
- **Historical Foundation:** Because data is inherently a measure of things that have already happened, analytics by definition must always examine historical data.

The Evolution of Analytics Terminology

- **A Modern Buzzword:** The term "analytics" rose to massive prominence around 2005, largely driven by the introduction of tools like Google Analytics.
- **Classic Roots:** However, the ideas and mathematics behind analytics have been around for more than 100 years.
- **Evolving Names:** Throughout the decades, this field has been called cybernetics, data analysis, neural networks, pattern recognition, statistics, data mining, and data science.

A Pragmatic Rise: Its recent rise is heavily pragmatic; as organizations collect massive amounts of data, they naturally progress toward using it to improve estimates and overall efficiency.

Understanding Business Intelligence (BI)

- **BI Focus:** Business intelligence (BI) is a vast field focused on summarizing interesting characteristics of existing data.
- **Key Outputs:** The typical outputs of BI analyses are reports or dashboards used by analysts and decision-makers.
- **Key Performance Indicators:** These tools frequently utilize Key Performance Indicators (KPIs) to help executives understand the state of the organization.
- **User-Driven Nature:** BI is fundamentally “user-driven,” meaning a domain expert decides exactly which predetermined metrics are interesting to report on.

Characteristics and Limitations of BI

- **Retrospective Questions:** BI excels at describing the past for a specific unit of analysis (like a customer or a transaction).
- **Examples in Fraud:** "How many cases were investigated last month?" or "How much revenue was recovered?"
- **Examples in Retail:** "Which regions had the highest response rates?" or "How many repeat purchasers were there?"

The Limitation:
While these reports tell you exactly what happened, they are not actionable decisions in and of themselves.

Defining Predictive Analytics

- **Definition:** Predictive analytics is the process of discovering interesting and meaningful patterns within data.
- **Data-Driven Methodology:** Unlike user-driven BI, predictive analytics is "data-driven," meaning algorithms derive the key characteristics of the models directly from the data itself.
- **Inductive Learning:** Instead of relying on human assumptions, powerful algorithms use induction to identify which variables, parameters, and coefficients belong in the model.

Multidisciplinary Field: It draws heavily from pattern recognition, statistics, machine learning, artificial intelligence, and data mining.

The Power of Algorithmic Automation

- **Automated Discovery:** Predictive algorithms automate the complex process of finding patterns, often discovering the very mathematical form of the models themselves.
- **Beyond Human Capacity:** Algorithms can test tens of thousands of potential combinations of inputs to identify the most predictive ones.
- **Example (Combinatorics):** To analyze a 3-way pattern among 50 variables, a human would have to manually examine 19,600 3D scatter plots.

Managing Complexity: Algorithms quickly sift through these combinations, reducing them down to a manageable subset of powerful predictors for the human analyst.

The Questions Predictive Analytics Asks

- **Prospective Focus:** Predictive analytics shifts the focus from simply summarizing the past to calculating the mathematical "likelihood" of a future event.
- **Examples in Fraud:** "What is the likelihood the transaction is fraudulent?" or "What is the expected amount of fraud?"
- **Examples in Retail:** "What is the likelihood an e-mail will be opened?" or "Which product is a customer most likely to purchase next?"

Data-Driven Selection: Algorithms test all possible patterns to see which ones actually predict this "likelihood" best.

Do Models Just State the Obvious?

- **The Common Criticism:** Decision-makers sometimes claim that predictive models simply confirm obvious relationships they already knew existed.
- **The Reality:** Models go much further; they identify the absolute best way to predict a target out of all possible variables and complex interactions.
- **Uncovering Complexity:** A human might guess that "length of residence" predicts a campaign response, but the model can prove that "number of contacts" is actually far more predictive.

Measuring Exact Impact: Models reveal not just which variables are predictive, but exactly how much better those combinations predict the target than individual attributes alone.

Comparison Table - BI vs. Predictive Analytics

This table highlights the fundamental operational differences between basic reporting and advanced analytics.

Feature	Business Intelligence (BI)	Predictive Analytics
Primary Focus	Retrospective (analyzes past performance)	Prospective (predicts future outcomes)
Key Output	Reports, dashboards, KPIs	Predictions, probabilities, risk scores
Approach	Query-driven (user asks questions)	Model-driven (algorithms discover patterns)
Example Question	How many fraud cases occurred last month?	What is the probability of fraud in this transaction?

The Role of Historical Data in Analytics

- **The Paradox of Forecasting:** Although we say predictive analytics "forecasts future events," models are built entirely using historical data.
- **Shifting the Timeline:** To predict the future, analysts must "roll back" the data to a time prior to the date the future event actually became known.
- **The Core Assumption:** The entire field rests on the assumption that future behavior will remain consistent with past behavior.

Real-World Application: If a mathematical pattern successfully predicted a purchase in the past, we assume that same pattern will predict future purchases.

Flowchart - The Analytics Data Timeline

To prevent future data from leaking into the model, the timeline must be strictly managed.

Historic Data — Attributes Computed — Data Models Built — Target Computed → Future Deployment

- **Attributes Computed:** Input data must be collected strictly prior to the date of the predicted event.
- **Target Computed:** The event you want to predict (e.g., response to a mailer) becomes known at a specific date.
- **Future Deployment:** The trained model is applied to new unseen data where outcomes are unknown.

Who Uses Predictive Analytics?

- **Historical Niche:** In the 1990s and early 2000s, advanced analytics was limited to highly forward-looking organizations with deep pockets.
- **Mainstream Adoption:** Today, analytics has evolved into an instrumental and required component for most mid- to large-sized organizations.
- **Universal Scalability:** Because of the explosion of data and processing power, even small for-profit and non-profit organizations now benefit heavily from predictive analytics using open-source software.

Summary and Takeaways

- **Core Definition:** Analytics uses computational methods and historic data to report influential patterns and affect critical decisions.
- **Data-Driven Discovery:** Predictive analytics automates the discovery of complex patterns using algorithmic induction rather than human assumptions.
- **Predictive Shift:** Moving beyond Business Intelligence requires a shift from user-driven retrospective reports to data-driven probabilistic forecasting.

The Shared Thread: Whether identifying marketing responders or guiding smart munitions, predictive modeling excels at estimating tremendously useful, unknown values.